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Professor Viswanatha Rao

ITAI 1371 Intro to Machine Learning

7 November 2026

# L10 Contribution Journal

Before starting the lab, I fixed the minor errors that had broken the notebook because every cell had collapsed onto one line, and the Task 3 cell was missing the seaborn import. I completed all three tasks: I used the elbow method to test k from 1 to 10, ran K-Means with k=4 and plotted the clusters, and used PCA to reduce the Iris data to 2 components that keep 97.77% of the variance. Then I checked each output against what I expected before I went to the next task. I wrote the Knowledge Check answers and then reviewed them against the Module 10 slides and earlier feedback from Professor Rao, so each answer names the key concepts from the module. I also wrote my reflection journal on what this lab taught me about unsupervised learning. I also submitted this lab for my group; I exported the notebook to PDF, collected each teammate's entries, and added them to the combined journals. Then I created the public GitHub repo, uploaded all three files, and submitted the URL in Canvas.

# Contribution Journal

ITAI 1371 | Lab 10: Unsupervised Learning | Student: Elquin Ponce

This journal describes my own work on Lab 10, written in my own words. Since the lab is graded individually, I am focusing on what I did and can verify myself.

## Code and analysis

- **Task 1 (elbow method):** I wrote and ran the loop that fits K-Means for k = 1 to 10, stores the inertia for each, and plots the curve. I used the plot to confirm that the bend is at k = 4.
- **Task 2 (K-Means clustering):** I fit K-Means with four clusters, pulled out the labels and cluster centers, and plotted the points colored by cluster with the centers marked.
- **Task 3 (PCA on Iris):** I reduced the four Iris features to two principal components, printed the explained variance (about 92.5% and 5.3%, 97.77% in total), and plotted the result by species.

## Problem solving

- I diagnosed a SyntaxError in the first code cell, which was caused by the notebook file losing its line breaks. I fixed it by rewriting the cell with proper line breaks.
- I noticed that the Task 3 plot used seaborn without importing it, and added the import so the cell would run.
- I ran all cells in order from the top and checked that every plot and printed output displayed correctly.

## Written work

- I answered the three Knowledge Check questions (supervised vs. unsupervised learning, why the largest k is not chosen from the elbow plot, and what over 95% explained variance says about the Iris data), using the actual numbers from my output.
- I drafted my reflective journal about what I learned, what I struggled with, and how this lab connects to my earlier work on scaling.

Aaliyah Phillips

11/7/2026

# Contribution Journal

For this week’s contribution, I worked through the Module 10 going over unsupervised learning. I worked through all the cells in the lab, answered the reflection questions in the end, and wrote a reflection journal page regarding the module. I made sure to turn in all documents to be turned in.

For this lab (Module 10 - Unsupervised Learning) I contributed to the group by independently working through the Notebook as usual, taking notes to ensure understanding, then having meaningful discussions with my group members about what we all learned. I also took the time to answer the knowledge check questions to the best of my ability, which I included in my reflection journal for submission.

# ITAI1371_L10 Contribution Journal_ Diane Lenhoff

## My Contributions

I worked on Lab 10, which focused on unsupervised learning using K-Means clustering and Principal Component Analysis (PCA). I ran the code, reviewed the graphs, and answered the knowledge check questions. I also worked through the elbow method to understand how to determine the best number of clusters and reviewed how the cluster centers were displayed on the graph.

## Challenges

One of my challenges was understanding the elbow method and why choosing more clusters does not necessarily give us a better model. I also had to review the PCA results to understand how reducing four features to two principal components could still explain 97.77% of the variance. I checked the plotting code to understand how `plt.show()` displays the graph.

## What I Learned

This lab helped me better understand how unsupervised learning identifies patterns without labeled data. I learned how K-Means groups similar data points and how PCA reduces the number of features while keeping most of the information. I can see how these methods could be useful when working with larger datasets, especially for identifying patterns and making the data easier to analyze.